

# Linbo Tang

☎ +1 608-886-7491 | ✉ linbo0204@outlook.com | 🔗 LinkedIn | 🐙 GitHub | 📁 Portfolio

## EDUCATION

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### Harvard University

*S.M in Data Science; GPA: 4.00/4.00*

MA, USA

*Sep 2024 – May 2026 (expected)*

### University of Wisconsin - Madison

*B.S in Computer Science and Data Science; GPA: 4.00/4.00*

WI, USA

*Sep 2020 – Dec 2023*

## RESEARCH INTERESTS

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Computer vision, generative modeling, and medical imaging — with a focus on integrating geometric, temporal, and physical constraints into deep learning frameworks for interpretable and domain-adaptive perception.

Broadly, I aim to develop generalized visual intelligence: computational models that can perceive, reason about, and reconstruct visual structure.

## RESEARCH EXPERIENCE

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### Zickler's Vision Lab

*Harvard University*

Dec 2024 – Present

*Advisor: Prof. [Todd Zickler](#)*

- Developing a diffusion-based framework for stereo vision that jointly models disparity and geometric scene structure from synthetic datasets.
- Adapted an **hourglass diffusion transformer** for stereo disparity estimation, employing image-conditioned geometric embeddings and progressive diffusion refinement to achieve higher depth fidelity and structural consistency.
- Developed a large-scale synthetic data generation pipeline producing stereo image pairs, disparity maps, and geometric annotations for model training, evaluation, and ablation studies.

### Psychiatry and Radiology Image Computing Group

*Harvard Medical School*

Sep 2024 – Present

*Advisor: Prof. [Yogesh Rath](#)*

- Developed a novel **scanner-adaptive, coil-level** denoising framework that conditions the network on measured noise statistics to achieve robust performance across scanners and protocols, submitted to **ISBI 2026**
- Designed a supervised training strategy with synthetic coil-wise noise realizations to ensure physically consistent noise modeling and improved generalization across acquisition settings.
- Introduced an attention-level conditioning mechanism via feature-wise modulation to dynamically adjust denoising strength according to local noise context.
- Evaluated the framework on in-vivo multi-coil diffusion MRI data, achieving up to a **23%** reduction in residual noise compared to the next best-performing method

### MSC Computer Vision Group

*University of Wisconsin - Madison*

Jan 2023 – Feb 2024

*Advisor: Prof. [Vikas Singh](#)*

- Curating **EmotionNet5k**, a novel selfie video dataset containing 5000 samples of facial expressions, with over 300 stars on [GitHub](#);
- Implementing a SOTA learnable wavelet model to remove co-founders and achieve a **3.13%** increase in accuracy on the downstream Facial Emotion Recognition (FER) task.
- Utilized **Category Theory** to improve compositionality of **StableDiffusion** by learning morphisms between embeddings to ensure reflexivity, transitivity, and neighborhood constraints;
- Developed metrics to verify Transitivity Axioms compliance and correlate with CLIP scores for model assessment.

### MIT CSAIL

*MIT EECS*

Apr 2025 – Present

*Advisor: [Dr. Giannis Daras](#) and [Prof. Jon Tamir](#)*

- Developed **ContextMRI**, a diffusion-based MRI reconstruction framework that leverages **fine-grained metadata embeddings** to condition on imaging parameters such as TR, TE, and flip angle.
- Designed a **structured embedding architecture** separating categorical and numerical attributes, using discrete embeddings and sinusoidal encodings respectively for improved quantitative representation.
- Introduced an **out-of-distribution (OOD) training strategy** by incorporating cardiac MRI data with distinct parameter ranges, forcing reliance on metadata priors and improving generalization.

- Demonstrated superior reconstruction accuracy and interpretability on the fastMRI brain dataset, particularly under diverse or uncommon acquisition settings.

## Ubisoft La Forge

*Research Student; La Forge*

Apr 2024 – Aug 2024

*Advisor: Alexis Rolland*

- Developed a generative base-material creation pipeline utilizing **StableDiffusion XL** and **inverse rendering** models to obtain rendered textures from text prompts and physically based rendering (PBR) maps from rendered textures.
- Refined a SwinIR-based super-resolution model for PBR maps by introducing a multi-branch architecture that separately processes base color, normal, roughness, and height maps while enforcing inter-map consistency.
- Achieved higher perceptual fidelity and improved structural coherence across PBR maps compared to standard single-branch and texture-domain super-resolution baselines.

## TEACH EXPERIENCE

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### Teaching Fellow

*Harvard University*

Aug 2025 – Present

*Advisor: Prof. Michael Brenner*

- Led weekly discussion sections for **AM 215: Mathematical Modeling**, covering topics such as dimensional analysis, Monte-Carlo simulation, and extreme-value statistics.
- Collaborated with the teaching staff on the design of homework assignments and course structure, providing individualized guidance to students throughout the semester.

### Undergraduate Teaching Assistant

*University of Wisconsin–Madison*

Sep 2021 – May 2023

*Advisor: Prof. [Frederic Sala](#)*

- Assisted in teaching four core courses: CS 200 (Programming I), CS 300 (Programming II), CS 400 (Programming III), and CS 540 (Introduction to Artificial Intelligence).
- Guided students through fundamental and intermediate programming concepts, data structures, and AI algorithms, emphasizing debugging and problem-solving strategies.
- Conducted tutoring sessions twice weekly, supporting over **400 students** across four semesters.
- Created and maintained a public [teaching portfolio](#) showcasing section materials, code examples, and instructional resources.

## PUBLICATIONS

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### SCANNER-ADAPTIVE COIL-LEVEL DENOISING FOR DIFFUSION MRI USING EXPLICIT NOISE PRIORS

**Linbo Tang\***, Qiang Liu\*, Lipeng Ning, Yogesh Rath

*ISBI 2026.*

## TECHNICAL SKILLS

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**Programming Languages:** Java, Python, C/C++, R, JavaScript, CSS, HTML, Kotlin;

**Frameworks:** Pytorch, Tensorflow, Scikit-Learn, MySQL, OpenCV, Flask, React, Selenium;

**Platforms:** Isaac Sim, UE4, AWS, GCP, Docker, Linux, Git.